nitibot.ai

Your Startup's Legal Co-Pilot.

PROJECT SCOPE DOCUMENT

NITI Bot — AI Legal Compliance for Startups

Your Startup's Legal Co-Pilot

HIT Hackathon 2026 • A4E Co-Founder Initiative

Project Name	NITI Bot	**Document Version**	v1.0
Document Date	09 April 2026	**Status**	Draft — Under Review
Project Owner	Team Lead (Samrat)	**Deadline**	05 May 2026
Prepared By	Samrat, Sujal, Sanskriti, Sonali, Ruchi, Zuha, Dipsubhra Bhunia	**Classification**	Confidential

1. Executive Summary

First-time startup founders in India face a critical knowledge gap when it comes to legal compliance. GST registration thresholds, MCA annual filing requirements, MSME registration benefits, and Startup India eligibility criteria are buried inside hundreds of pages of government documents written in dense legal language. The cost of missing a filing deadline or misunderstanding a threshold ranges from monetary penalties to complete business shutdown.

This project delivers an AI-powered compliance assistant — NITI Bot — that enables Indian startup founders to ask plain-English questions and receive accurate, cited, actionable answers grounded in official government regulations. The bot uses Retrieval-Augmented Generation (RAG) to ensure every response is traceable to a specific regulation section, eliminating hallucination risk common in general-purpose AI tools.

> Strategic Positioning: NITI Bot is India's first regulation-grounded, citation-enforced compliance assistant designed specifically for first-time startup founders. Unlike general AI tools, it refuses to answer without a source document, making it uniquely trustworthy.

2. Project Objectives

2.1 Primary Objectives

- Build a functional RAG chatbot trained on GST Act, MCA guidelines, MSME Act, and Startup India Handbook.
- Enable plain-English compliance queries with cited, checklist-format responses.

- Deploy a publicly accessible live URL for judging and user access by 05 May 2026.
- Demonstrate a working demo with at least five pre-validated founder questions.

2.2 Secondary Objectives

- Establish a scalable document pipeline that can absorb additional regulations post-hackathon.
- Produce a pitch-ready business model and post-hackathon roadmap for incubator evaluation.
- Position the product for Startup India recognition and potential B2B incubator licensing.

3. Scope of Work

3.1 In Scope

Work Item	Description
RAG Pipeline	Flowise-based retrieval pipeline with ChromaDB vector store; 4 government documents chunked at 512 tokens with 64-token overlap
Document Corpus	GST Act summary (CBIC), MCA21 filing guide, MSME Act 2006, Startup India Handbook — all sourced from official government portals
AI Generation Layer	Claude API (Sonnet) for response generation with a citation-enforcing system prompt that refuses to answer without a source chunk
Chat Interface	React-based or Flowise-embedded web chat UI with topic selector (GST / MCA / MSME & Startup India) and citation display panel
Deployment	Self-hosted Flowise on Render or Railway (free tier); ngrok tunnel as fallback for demo day
Testing	30+ real founder question test bank with pass/fail citation accuracy tracking
Demo Package	5 scripted demo questions, 90-second screen recording, 8-slide pitch deck
Documentation	Project Scope Document, technical architecture notes, team Gantt chart

3.2 Out of Scope

Note: The following items are explicitly excluded from the MVP build. They are acknowledged as roadmap items but will not be delivered before the May 5 deadline.

- User authentication, login, or session management
- Document upload by end users
- WhatsApp / Telegram / SMS integration
- Automated compliance deadline reminders
- Multi-language support (Hindi, Bengali, etc.)
- Lawyer or CA referral network
- Mobile application (iOS / Android)
- Payment gateway or subscription billing

4. Team Structure & Responsibilities

Member	Role	Core Skills	Primary Phase
Samrat	UI/UX Lead & App Dev	Java, Kotlin, C++, App Dev, UI/UX	Phase 4 — Frontend UI
Sujal	ML & Backend Engineer	Java, Python, C, C++, ML, AWS Basics	Phase 2 & 3 — RAG & Testing
Sanskriti	ML & Data Engineer	Python, ML Model Dev, AWS, Data Handling	Phase 2 & 3 — Embedding & Test
Sonali	Web Developer	Java, C, HTML, CSS, JS, Web Development	Phase 4 — Web Frontend
Ruchi	ML & Python Engineer	Java, Python, ML, HTML, CSS, AWS Basics	Phase 3 — Prompt & Testing
Zuha	Fullstack Lead & DevOps	Java, C, Python, ML, AWS, Fullstack	Phase 5 — Deployment & Infra
Dipsubhra Bhunia	Communications & Pitch Lead	Strong Communication, Presentation, Strategy	Phase 6 & 7 — Demo & Pitch

5. Technical Architecture

5.1 Technology Stack

Layer	Tool / Service	Purpose
Orchestration	Flowise (self-hosted)	Visual RAG pipeline builder; manages retrieval, prompt chaining, and Claude API calls
Vector Store	ChromaDB	Stores document chunk embeddings; performs semantic similarity search at query time
LLM (Generation)	Claude API — Sonnet model	Generates cited, checklist-formatted compliance answers from retrieved regulation chunks
Frontend	React + TailwindCSS or Flowise Embed	Chat interface with topic tabs, citation display, and pre-filled demo questions
Document Sources	Government PDFs (4 documents)	GST Act, MCA21 Guide, MSME Act 2006, Startup India Handbook — all public domain
Deployment	Render / Railway (free tier)	Self-hosted Flowise instance; ngrok fallback for demo day if cloud setup fails
Embeddings	Flowise built-in embedder	Converts 512-token document chunks into vector representations for ChromaDB

5.2 Data Flow

The system follows a standard RAG (Retrieval-Augmented Generation) pattern with citation enforcement:

- User types a plain-English compliance question into the chat interface.

- The query is embedded into a vector and sent to ChromaDB for similarity search.
- The top 3-5 most relevant document chunks are retrieved, each tagged with their source section metadata.
- The system prompt, retrieved chunks, and user query are assembled and sent to the Claude API.
- Claude generates a checklist-format answer using only the provided context, appending the source section citation.
- The frontend displays the answer with the citation panel highlighted below the response.

5.3 Claude System Prompt Design

> System Prompt (Core Instruction — enforced at every query):
>
> You are a compliance assistant for Indian startup founders. Answer ONLY from the provided context documents. Format all answers as numbered action checklists. Always cite the exact regulation section at the end of your response. If the context does not contain a definitive answer, respond: 'This requires a qualified CA or lawyer. Here is what the regulation says generally.' Never invent penalties, thresholds, or deadlines not present in the retrieved chunks.

6. Deliverables

#	Deliverable	Target Date	Owner
D1	Flowise RAG pipeline with ChromaDB integration	14 Apr 2026	Zuha, Sujal
D2	4 government PDFs chunked, embedded, and tested	12 Apr 2026	Sanskriti, Ruchi
D3	Citation-enforcing Claude system prompt (v1)	14 Apr 2026	Sujal, Ruchi
D4	30-question test bank with retrieval accuracy report	19 Apr 2026	Sanskriti, Sujal
D5	React chat UI with topic tabs and citation panel	26 Apr 2026	Samrat, Sonali
D6	Live deployed URL (Render / Railway)	29 Apr 2026	Zuha
D7	90-second screen demo recording	01 May 2026	Dipsubhra Bhunia, Samrat
D8	8-slide investor/judge pitch deck	02 May 2026	Dipsubhra Bhunia, Ruchi
D9	Project Scope Document (this document)	09 Apr 2026	All Members
D10	Final rehearsed pitch + judge Q&A prep	04 May 2026	Dipsubhra Bhunia

7. Project Milestones & Timeline

Milestone	Date Range	Completion Date	Status
M1: Environment Setup	09 – 10 Apr	10 Apr 2026	**In Progress**
M2: RAG Pipeline Live	11 – 14 Apr	14 Apr 2026	Planned
M3: Testing & Tuning Complete	15 – 19 Apr	19 Apr 2026	Planned
M4: Frontend UI Done	20 – 26 Apr	26 Apr 2026	Planned
M5: Live Deployment	27 – 29 Apr	29 Apr 2026	Planned
M6: Demo & Pitch Package	30 Apr – 02 May	02 May 2026	Planned
M7: Final Rehearsal	03 – 04 May	04 May 2026	Planned
M8: SUBMISSION DEADLINE	05 May 2026	05 May 2026	**TARGET**

8. Assumptions & Constraints

8.1 Assumptions

- All team members have access to a personal laptop with at least 8GB RAM for local Flowise development.
- Claude API free tier or a shared trial API key will be available for the duration of the project.
- Government PDF documents are freely available and in the public domain (GST Act, MCA, MSME Act, Startup India).
- Render or Railway free-tier deployment will sustain demo-level traffic (< 50 concurrent users).
- Team availability: all members commit a minimum of 4 hours per day from April 9 to May 5.
- Flowise self-hosted installation does not require a paid licence for hackathon use.

8.2 Constraints

- Budget: Zero rupees allocated. All tools must be free or open-source.
- Timeline: 27 calendar days, non-negotiable. No scope additions after April 20.
- Team Skill Boundary: No dedicated full-time DevOps engineer. Zuha handles deployment alongside ML work.
- API Cost: Claude API usage must remain within free or trial tier limits during development.
- Legal: The product must include a clear disclaimer that it does not constitute legal advice.
- Scope Freeze: No new features after April 20 (Day 12). All effort shifts to polish and pitch prep.

9. Risk Register

Risk	Probability	Impact	Mitigation Strategy

RAG retrieval returns wrong regulation chunks	**Medium**	**High**	Test with 30 questions before demo. Tune chunk size (512 tokens) and overlap (64 tokens). Add re-ranking if Flowise supports it.
Claude API generates hallucinated legal info	**Low**	**Critical**	System prompt enforces: answer only from retrieved context. Bot responds 'Please consult a CA' when context is insufficient.
Live demo fails at the event (API/network down)	**Medium**	**High**	Pre-record a 90-second screen demo. If live demo fails, walk judges through the recording — fully acceptable at hackathons.
Government PDFs are outdated or amended	**Low**	**Medium**	Tag each document chunk with version/year metadata. Citation display shows the source date so founders can verify currency.
Deployment fails on Render/Railway	**Low**	**High**	Maintain a local ngrok tunnel as a primary backup. Test deployment on Day 5 (April 27) — 8 days before deadline.
Team member unavailability (illness/exams)	**Medium**	**Medium**	Critical path tasks (RAG pipeline, deployment) are owned by Zuha and Sujal with documented backups for each step.
Scope creep beyond MVP	**High**	**Medium**	Scope freeze enforced after April 20. Product Owner (Zuha) has authority to reject any feature additions after that date.
Legal liability concerns from judges/investors	**Medium**	**Low**	Prominent disclaimer on every page: 'For informational purposes only. Not legal advice.' Modelled on ClearTax and Vakilsearch disclaimers.

10. Business Model & Market Opportunity

10.1 Market Sizing

Total Addressable Market: ~14 million first-time founders and MSMEs registered in India (MCA data, 2024)

Serviceable Market: ~2 million tech-savvy startup founders with internet access who cannot afford CA retainers

Initial Target: Student founders, solopreneurs, and early-stage startups in incubators and accelerators

10.2 Revenue Phases

Phase	Model	Target Segment	Pricing	Timeline
Phase 1	Freemium	Student founders, solopreneurs	Free (10 Q/mo) + Rs.299/mo Pro	M 1-6
Phase 2	B2B Licensing	Incubators, IIT/NIT cells, TiE	Rs.5,000-20,000/mo per org	M 6-12
Phase 3	CA Referral	High-complexity queries	15-20% referral commission	Year 2
Phase 4	Compliance SaaS	Growth-stage startups	Rs.999/mo compliance manager	Year 2+

11. Success Criteria

11.1 Hackathon MVP Success Criteria

Criterion	Measurement	Threshold
RAG pipeline answers founder questions	Live demo with 5 scripted questions	**100% success rate**
Citation shown on every response	Manual check of 10 test responses	**10/10 responses cited**
Bot refuses to hallucinate	Test 5 out-of-scope questions	**0 hallucinated answers**
Deployed on public live URL	URL accessible from judge's device	**Live by Apr 29**
Demo completes in under 90 seconds	Timed rehearsal run-through	**Under 90 seconds**
Pitch deck covers all 8 required slides	Peer review against slide outline	**8/8 slides present**
Team can answer 5 standard judge questions	Mock judge Q&A session on May 4	**All 5 answered clearly**

12. Document Approval & Sign-Off

This document requires review and acknowledgement by all core team members before development begins. By signing below, each member confirms they have read, understood, and agreed to the scope, responsibilities, and timeline defined in this document.

Name	Role	Signature	Date
Samrat	UI/UX Lead & App Dev		
Sujal	ML & Backend Eng.		
Sanskriti	ML & Data Eng.		
Sonali	Web Developer		
Ruchi	ML & Python Eng.		
Zuha	Fullstack Lead & DevOps		
Dipsubhra Bhunia	Pitch & Communications Lead		

> Document Control: This is a living document. Changes after April 20, 2026 require written approval from the Project Owner (Zuha) and must be logged with date, author, and reason for change. Scope additions after April 20 are not permitted without explicit team consensus.

— End of Document —